

CROIP - Approach 2: Three-System Model

CROIP and its ratepayer app are the platform. Concessionaires build and run municipal systems on CROIP's APIs.

1. Summary

CROIP owns the property registry, ratepayer accounts, rate cards, bill calculation and all payment channels, and provides the universal ratepayer app. Concessionaires are companies engaged by municipalities to build and run office and agent apps on CROIP's APIs. They verify properties and request bills; they do not collect payments or hold ratepayer payment relationships. Ratepayers have one global login and see every property they own, across every municipality.

2. Parties and scope

Party	Role
CROIP	Builds the platform and the universal ratepayer app. Provides the property registry (polygons, municipality tag), ratepayer accounts, payment APIs routed to a partner bank platform, rate cards and bill calculator, payment log, reconciliation, ticket routing, developer portal and sandbox.
Municipality	The authority behind rates. Engages a concessionaire. Its approved person confirms the concessionaire's appointment and supplies the yearly rate card.
Concessionaire	Company engaged by a municipality. Builds and runs its own office app and agent app on CROIP's APIs. Contracts only with CROIP. May serve one or more municipalities, but each municipality has exactly one concessionaire.
Ratepayer	One global login. Adds, claims and views properties, sees bills, pays through CROIP, raises disputes and support requests.
LVD (external)	Land Valuation Division of the Lands Commission. Prepares valuation lists and rateable values. CROIP exchanges property and valuation data with it.

3. Data ownership and trust boundary

- All properties live in CROIP, tagged by municipality. A property belongs to exactly one municipality.
- A concessionaire sees only the municipalities in its scope. Its staff and agents are scoped per municipality as well.
- Ratepayers are global: their account and property list are not tied to any one concessionaire.
- Property and ratepayer data belongs to CROIP and the municipality, not the concessionaire. A municipality can switch concessionaires without migrating data.
- Identity and reference design (including Ghana Card linking) is parked. Until then, ratepayers can start with a CROIP-issued ID and phone or email login.

4. Property verification flow

- 1 A ratepayer adds a property in the app (or the concessionaire registers one). CROIP checks for duplicates and overlaps and routes it to the municipality's concessionaire. The ratepayer sees a status tracker.
- 2 An office staffer assigns the property to one agent. Each property goes to a single agent; a municipality can run many agents.

- 3 The agent visits and captures a boundary polygon of at least 4 points, photos, ownership evidence, and the inputs that select the rate: class or zone, usage type, development status (completed, uncompleted, undeveloped) and number of storeys.
- 4 The submission goes to **review** and is never auto-verified. The staffer who made the assignment approves it or sends it back.
- 5 The approved record, with polygon, is sent to CROIP and the ratepayer sees the property as verified. Until approval it is not shown as the ratepayer's and cannot be billed.

5. Valuation and billing

- **LVD sets the value.** The rateable value comes from LVD's valuation list or an LVD-appointed valuer. Agents capture inputs only; they do not set values.
- **Two ways to bill.** Assessed property: rateable value x the assembly's rate, subject to its minimum. Unassessed property: a flat provisional rate by class and use. Assemblies also publish rates for uncompleted blocks and undeveloped plots, and rebates. This follows published fee-fixing resolutions; confirm the reconciliation rules with LVD and the assemblies.
- **Rate cards.** One per municipality, versioned by year with effective dates. Never edited while live; mid-year changes become a new version.
- **CROIP calculates the bill.** The concessionaire requests a bill for a property and year; CROIP computes the amount from the property record, rate card and rateable value. Each bill records the rate-card version used. Adjustments need a reason code and approval.
- **Rate-card process.** The municipality's approved person provides the rate card. CROIP checks it against the published fee-fixing resolution and confirms it with the approved person. One CROIP person enters it and a second approves it. Sample properties are tested before it goes live, and the signed document is kept with that version.
- **Disputes.** Boundary and ownership disputes go to the concessionaire, which can send an agent to recheck. Valuation disputes are routed to LVD's process.

6. Payments and money flow

- Payment channels (banks, mobile money, cards, POS, web, USSD, agents) sit inside CROIP as one unified set of payment APIs.
- CROIP acts as a middleman for a bank's payment platform. It is not a fintech, but it keeps full log details of every payment.
- Design principle: CROIP orchestrates payments into the bank's collection or settlement accounts and does not hold balances. Use the bank's hosted checkout or tokenization so card data never touches CROIP.
- Funds and settlement are kept separate per municipality. Field agents and POS devices use the same payment APIs.

7. Contracting and onboarding

Layer	What happens
Per concessionaire (once)	Contract with CROIP and acceptance of API terms. Developer portal access and keys. Pass the sandbox conformance suite, which is the go-live gate.
Per municipality (each one added)	Appointment confirmed by the municipality's approved person. Rate card verified and entered. Configuration test before go-live. CROIP removes a municipality from a concessionaire's scope on written notice from that approved person.

Responsibility. The concessionaire is accountable to CROIP for the accuracy of the data it captures, the bills it requests, the conduct of its agents and staff, the quality and uptime of its apps, and first-line resolution of municipality-specific issues within agreed service levels. CROIP is accountable for platform uptime, payments and ticketing. Enforcement levers: published scorecards, API throttling or suspension for serious breaches, and settlement offsets for refunds and penalties (needs legal input).

8. Support

- **Single front door:** in-app help, one shared number or WhatsApp line, and USSD. Every request becomes a CROIP ticket carrying the ratepayer, property, bill and municipality, and routes automatically.
- **CROIP handles:** general "how it works" questions (including from people with no property yet), login and account problems, payment failures, "paid but not showing", receipts and refunds.
- **The concessionaire handles:** anything tied to a property or municipality, such as bill queries, boundary disputes, registration requests and rechecks.
- Tickets can be reassigned between CROIP and the concessionaire without the ratepayer repeating their story. Status is visible in the app, and response times are reported.

9. The API as a product

- Published OpenAPI specification and event schemas, with a versioning and deprecation policy.
- Developer portal with documentation, keys and a sandbox that uses synthetic data only.
- Per-municipality scoped credentials, full audit logs, and the ability to suspend a concessionaire's keys. Confirm data protection obligations for third-party access.
- Webhooks for bill and payment events, with documented retries and idempotency.
- Conformance test suite that concessionaires run themselves.

10. Risks and mitigations

Risk	Mitigation
CROIP is a single point of failure for all collection.	High-availability design, queued payment retries, status page.
Someone claims or adds a property that is not theirs, or floods the queue.	Claim-and-verify before ownership shows; duplicate and overlap checks; minimum details and limits on open requests.
Regulatory exposure from routing money.	No balances held; confirm licensing position with legal counsel and the partner bank.
Quality varies across concessionaire systems.	Conformance suite as go-live gate, shared ticketing, scorecards, and one consistent ratepayer app.
Wrong bills from manual rate entry.	CROIP calculates from versioned, verified rate cards.

11. Open items to resolve

- Identity and reference service design, to be agreed with all stakeholders.
- Settlement split: who receives funds and how they divide, per municipality.
- Billing and payment status while a property is under dispute.
- Legal and licensing position on CROIP orchestrating payments, and on any per-transaction fee.
- LVD data exchange (digital or manual) and whether provisional payments are credited against the final assessment.
- Who staffs first-line support: a shared CROIP team, or concessionaire teams working from CROIP's ticketing.
- Review-queue rules, such as reassignment when a reviewer is unavailable.
- API pricing: onboarding fee, per-municipality fee, or per-transaction fee.